BaatSeBharat Project Roadmap

Research & Product Roadmap: BaatSeBharat

We are building a **publication-standard** system by merging the "Hardcore" research requirements with a functional product prototype.

Part 1: The "Hardcore" Product Prototype (Weeks 510)

Goal: Build a functional, interactive Streamlit tool that implements advanced NLP and financial metrics.

1. Advanced Neural Topic Engine (Weeks 5-7)

- **Hybrid Architecture**: Transition from single-model (NMF/LDA) to a **Hybrid Neural-Probabilistic** approach using **IndicBERT/SBERT** embeddings with **Contextualized Topic Models (CTM)** and **ProdLDA**.
- **Multi-Granularity Modeling**: Segment transcripts at **Sentence**, **Paragraph**, and **Episode** levels. Implement hierarchical aggregation to reduce topic bleeding.
- **Zero-Shot Automatic Labeling**: Interface with **BART/DeBERTa** models for bias-free, autonomous topic categorization.
- **Sentiment Overlay**: Integrate **FinBERT** for domain-specific sentiment (Optimism vs. Caution).

2. Live Market Intelligence & Robustness (Weeks 8-10)

- **Predictive Validation**: Use topic distributions as features to predict **Sectoral Volatility (Risk)** and **Abnormal Returns**.
- **Weighted Moment Analysis**: Implement **Probability-Weighted Moments (PWM)** to quantify the "shock impact" of specific speeches on market tails (extreme fluctuations).
- **Topic Stability Dashboard**: Display **Jaccard Similarity** and **Word Movers Distance (WMD)** in the UI to prove topic consistency across random seeds.
- **Temporal Topic Alignment**: Build an alignment engine (Hungarian Matching) to track how topics evolve over a 10-year span.

Part 2: The Journal Paper (Weeks 1116)

Goal: Formalize the approved prototype into a publication-standard research paper targeting high-impact venues (e.g., IEEE Access, Expert Systems with Applications).

1. Academic Rigor & Novelty (Weeks 11-13)

- **Causal-ish Validation (Ablation Studies)**: Systematically test the "predictive lift" added by POS filtering, SBERT embeddings, and Dynamic Alignment.
- **Ensemble Consensus**: Formalize the aggregation of LDA/NMF/CTM results into a single

"model-agnostic" topic set.

- **Multimetric Evaluation**: Report C_v, NPMI, UMass, and Diversity scores for validation.

2. Publication & Refinement (Weeks 14-16)

- **Methodology Formalization**: Translate the prototype logic (e.g., PWM calculations, CTM loss functions) into mathematical formalisms.
- **High-Res Visualization**: Export publication-quality charts (Heatmaps, Temporal Flow Diagrams, Stability Plots).

Technical Stack for Prototype

Component	Technology
Embeddings	Sentence-BERT / IndicBERT
Topic Model	CTM / ProdLDA / Ensemble Consensus
Sentiment	FinBERT (for financial tone)
Risk Metrics	Probability-Weighted Moments (PWM)
Stock Data	yfinance (Tick-level / Daily resolution)
Validation	Granger Causality + Ablation Tables

Next Steps

1. **Approval**: Does this roadmap align with your academic goals?
2. **Initial Task**: I recommend we start by upgrading the `requirements.txt` and adding a basic **Sentiment Analysis** feature to the Topic Modeling page.